

Kundai Farai Sachikonye
Auf der Lichtung 19, 82178 Puchheim, Germany
github.com/fullscreen-triangle
kundai.sachikonye@bitspark.com
(+49) 1626386344

18 May 2026

Forschungszentrum Jülich GmbH
HDS-LEE Graduate School
52428 Jülich, Germany
Ref. 2025D-179

Re: PhD Position — Probabilistic Metabolic Flux Analysis (Ref. 2025D-179)

Dear Hiring Committee,

I am applying for the doctoral position in probabilistic metabolic flux analysis within the HDS-LEE graduate school (Ref. 2025D-179). The position's core challenge — developing a Bayesian inference framework robust to model misspecification for hybrid model- and data-driven metabolism modelling — sits at the intersection where I have worked most intensively: hierarchical Bayesian models, differentiable probabilistic inference, and large-scale biological data, with the specific application domain being metabolic measurement data from my doctoral research in Computational Lipidomics at TUM.

Bayesian inference, hierarchical models, and model misspecification. The fundamental difficulty in metabolic flux analysis (MFA) is that the stoichiometric model $\dot{\mathbf{x}} = S\mathbf{v}$ is always misspecified: unmeasured fluxes, unknown regulatory interactions, compartment-averaging artefacts, and isotope label dilution all introduce systematic deviations between model and data that point estimates cannot detect. A Bayesian hierarchical approach handles misspecification by placing priors over model parameters *and* over the degree of model inadequacy: the posterior distribution over flux vectors $p(\mathbf{v} \mid \mathbf{m})$ captures both estimation uncertainty and structural uncertainty jointly.

I have built and applied hierarchical Bayesian models in two directly relevant contexts. The Gospel genomic analysis framework implements Bayesian network inference for multi-omics data integration, using variational Bayes for posterior approximation in models with $\mathcal{O}(10^6)$ latent variables. The Syndrome disease modelling framework uses Bayesian $P(R \mid Q)$ — the posterior probability of answer quality given query class — as a gating mechanism over an eight-dimensional latent state, with continuous shrinkage priors over the coherence indices $\eta_i \in [0, 1]$. For the MFA context, these translate directly: the latent flux vector $\mathbf{v}$ plays the role of the disease state $\mathbf{D}$; the isotopomer balance equations play the role of the stoichiometric network; and the hierarchical prior over flux magnitudes encodes the biological constraints (thermodynamic feasibility, capacity constraints, mass balance) analogously to the sparse ℓ_1 regularisation I use for minimum intervention selection.

Beyond methodology, I have a theoretical result directly applicable to the model misspecification problem. The **Backward Training Principle** (developed in the Purpose framework) proves that a model calibrated at the worst-case misspecification scenario —

the hardest, most constrained instance of the inference problem — is calibrated at all less-misspecified instances by feasibility inclusion:

$$\mathcal{C}_{\text{hard}} \subseteq \mathcal{C}_{\text{easy}} \implies \epsilon\text{-competence on } \mathcal{C}_{\text{hard}} \implies \epsilon\text{-competence on } \mathcal{C}_{\text{easy}}.$$

For a Bayesian MFA framework, the worst-case misspecification is the fully unidentified model (maximum parameter uncertainty, minimum data information); calibration there propagates to all better-identified sub-models. A Forward Training Collapse Theorem establishes a positive Vapnik-Chervonenkis failure gap for the conventional easy-to-hard approach; a Sample Complexity Theorem gives $(N+1)\times$ savings for a cascade of depth N . This is a formal contribution to the project’s goal of designing hierarchical models that capture misspecification robustly.

Differentiable programming and HPC-optimised sampling. The project calls for differentiable, scalable inference algorithms using automatic differentiation and HPC-optimised sampling in Python and C++. My implementations use PyTorch as the primary differentiable computing substrate: every model in my frameworks is built on `torch.autograd` so that gradients through the full model — including through stochastic nodes via the reparameterisation trick — are available for gradient-based MCMC (NUTS, HMC) and variational inference (ELBO optimisation). For HPC scale, I implement distributed sampling via Ray: multiple Markov chains run in parallel across cluster nodes, with shared memory for the likelihood evaluation and asynchronous parameter collection. For performance-critical likelihood evaluations, I implement in Rust and expose to Python via PyO3 bindings, achieving 15–25 $\times$ speedup over pure Python for the inner loops of MCMC sampling. The Four-Sided-Triangle framework demonstrates this architecture in a deployed system: 8-stage inference pipeline with Rust core, Ray-distributed parallel execution, and Python orchestration layer.

Metabolic measurement data and flux analysis background. My doctoral research at TUM / Bayerisches Forschungsinstitut (2019–2021) was in Computational Lipidomics: mass spectrometry-based quantification of lipid species across hundreds of samples from distributed clinical sites. Lipidomics is the downstream measurement of lipid metabolic fluxes — the lipid species abundances in a sample are the integrated outputs of the synthesis, modification, transport, and degradation reactions in the lipid metabolic network. The pipeline I worked on covered: peak detection and alignment across mzML files >100 GB, ion annotation against LIPID MAPS and LipidBlast, multi-omics integration combining lipidomics with clinical covariates, and federated statistical modelling across distributed sites. This gives me direct familiarity with the data structures and measurement uncertainties (isotopologue distributions, matrix effects, ion suppression) that a probabilistic MFA framework must handle. The Lavoisier framework I subsequently built automates this pipeline at production scale: dual numerical and CNN-based spectral analysis, 98.9% NIST accuracy, Ray-distributed for HPC, >100 GB mzML support. Isotopically labelled metabolomics data (the primary input to ^{13}C -MFA) is structurally identical to the lipidomics data I have processed: the difference is the labelling scheme applied to the sample, not the measurement instrument or data format.

Open-source contributions and collaborative research. All frameworks I have built are publicly available on GitHub with documentation, validation reports, and reproducibility instructions. I contribute to open-source scientific computing projects and have implemented library-level components (Rust statistical kernels, Python API wrappers

for genomic databases, WebGPU compute shaders for physical simulation) that are used in the deployed applications.

Background. I hold an MSc in Computational Biology (Universität Tübingen) and conducted doctoral research in Computational Lipidomics at TUM. I work primarily in Python and Rust; C experience from Fraunhofer IGB and scientific computing contexts. I am legally resident in Germany with full work authorisation and available immediately; willing to relocate to Jülich. English is native; German is conversational.

Yours sincerely,

Kundai Farai Sachikonye